



Mockup of: SMS Spam Detection System — PRD v1.0

Real-time SMS spam & phishing defense — **classified before delivery.**

An NLP-powered classification API scores every inbound SMS as **ham** or **spam** in under 100ms using a Decision Tree model trained on TF-IDF features — quarantining malicious messages before they reach a phone, while a feedback loop and monthly retraining pipeline keep it adapting to new spam vocabulary.

96.59%

Test accuracy
(Decision Tree)

93.50%

Test precision on
spam class

<100ms

P99 classification
latency

13.5%

Spam rate across
SMS traffic

Who sees what

Four personas from the PRD's user stories, each with their own corner of the product.



CISO / Security Manager

Wants visibility into the SMS threat landscape and confidence the filter is working. — Story 1, 4

Dashboard · Alerts · Live Feed



Business Operations Manager

Needs a blocked legitimate message released fast, and a way to flag mistakes. — Story 2

Report a Misclassification · Quarantine



Telecom / Platform Engineer

Integrates the classification API into routing infrastructure and watches its health. —

Story 3

[Dashboard](#) · [Settings \(API keys\)](#)



ML / Review Analyst

Labels borderline messages within SLA and oversees model retraining and rollouts. —

Flow 5

[Review Queue](#) · [Model & Retraining](#)

Flow map

The five user flows from PRD §3.2 — each maps to one or more screens below.

FLOW 1 — REAL-TIME CLASSIFICATION

SMS in → preprocess → TF-IDF → Decision Tree → label + confidence → quarantine or deliver → audit log. <100ms

[Watch the live feed](#)

FLOW 2 — MISCLASSIFICATION FEEDBACK LOOP

User flags a wrong call → feedback record created → analyst confirms → correction queued for retraining.

[Open the report flow](#)

FLOW 3 — MONTHLY MODEL RETRAINING

Pull feedback + new messages → retrain → evaluate vs. holdout → promote if precision $\geq 93\%$ & recall $\geq 90\%$, else hold.

[See the retraining pipeline](#)

FLOW 4 — ADMIN MONITORING & ALERTING

Dashboard tracks spam rate, false-positive rate, accuracy, latency → alerts on spikes or degradation → weekly report.

[View the dashboard](#)

FLOW 5 — HUMAN-IN-THE-LOOP REVIEW (LOW CONFIDENCE)

Messages scored 40–60% confidence are routed to an analyst queue (not auto-blocked) → reviewed within 4 business hours → label applied and added to the training queue immediately.

[Open the review queue](#)

All screens

Ordered like a storyboard — open `index.html` any time to come back here.

01 **Sign in**

Security-console authentication for staff users.

02 **Security dashboard**

North-star metrics, spam-rate trend, live alert banner. FR7, Story 4

03 **Live classification feed**

Streaming view of every message scored in real time. Flow 1, FR1, FR2

04 **Quarantine queue**

Held spam, 30-day retention, one-click release. FR3, FR4, Story 2

05 **Review queue (uncertain)**

40–60% confidence messages awaiting an analyst label. Flow 5, FR6

06 **Message detail & explainability**

Hash, audit trail, and the top contributing TF-IDF terms. §4.4 Explainability

07 **Report a misclassification**

End-user portal for flagging a wrongly blocked message. Flow 2, Story 2

08 **Alerts & campaigns**

Spike detection, weekly security report, CSV export. FR7, Story 4

09 **Model & retraining**

4-model comparison, accuracy gate, version history. Flow 3, FR5, §5.1

10 **Settings & governance**

Thresholds, sender allowlist, API keys — gated by approval. §4.3

Static, click-through prototype — no backend. See [README.md](#) for assumptions, open questions, and PRD traceability.

Generated from SMS-SpamClassifier-PRD.pdf · v1.0



ShieldLine

SMS Threat Defense Console

Sign in to your console

Enter your work email to access the security dashboard, quarantine, and review tools.

Work email

priya.nandakumar@acmemobile.com

Password

[Forgot password?](#)

.....



☒ Keep me signed in on this device

Sign in →

or



Continue with company SSO

Access is limited to staff with a security, IT, or operations role at Acme Mobile. Need an account or seeing a wrongly blocked message instead? [Report a misclassification](#) →



Security dashboard

Real-time view of SMS classification across Acme Mobile

Last 24 hours

Export CSV



Possible spam campaign detected

Spam rate is **+24%** above the 7-day rolling average as of 14:32 today — this crosses the FR7 alert threshold (>20%). Two related alerts are open.

View alerts



Spam intercepted today

1,842

↑ 12% vs. yesterday



Spam rate

16.2%

North Star target ≥ 95% recall



False positive rate

5.8%

Target ≤ 6.5% · within range



Model accuracy

96.6%

≥ 95% maintained · v2.4



P99 latency

87ms

Target < 100ms · healthy

Spam rate — last 7 days

7-day rolling average: 13.1%

Spike on Sun

Mon

Tue

Wed

Thu

Fri

Sat

Sun

11.8%

16.8%

Confidence distribution

Messages classified today, by model confidence

0–20% (confident ham)

412

20–40%

1,205

40–60% — uncertain (Flow 5)

38



Top spam phrases this week

Highest-weight TF-IDF terms behind today's classifications

FREE	482
WIN / WINNER	391
CLAIM	358
PRIZE	312
URGENT	276
CASH	241

Recent activity

Last messages scored by the classifier

[Open live feed →](#)

SPAM	"WINNER!! You have been selected to receive a £900 prize reward. To claim call 09061..."	98.4%	14:37:02
HAM	"Reminder: Team standup moved to 9:45am tomorrow, same room as last week."	3.1%	14:36:54
UNCERTAIN	"Your order #48217 has shipped — track it here: trk.acme-deliveries.co/4q1z"	52.0%	14:36:41
SPAM	"URGENT! Your Mobile No. has won £2000 Bonus Caller Prize. Call 09066362231 ASAP!"	96.7%	14:36:29
HAM	"Can you pick up some milk on your way back? Running low, thanks!"	1.8%	14:36:15



Live classification feed

Flow 1 · every inbound SMS, scored and routed in < 100ms

Streaming live

AllSpamHamUncertain

Search by message hash...

TIME	MESSAGE PREVIEW	SENDER ID	LABEL	CONFIDENCE	EFFICIENCY	MESSAGE HASH (SHA-256)
15:32:23	"WINNER!! As a valued custo...	PRZ-WIN	SPAM	98.4%	30ms	sha256:85d89...
15:32:23	"Congratulations! You have b...	+1 302 555 0...	SPAM	91.8%	85ms	sha256:67212...
15:32:23	"Reminder: Team standup mo...	+44 7911 223...	HAM	3.1%	31ms	sha256:4d157...
15:32:23	"Nah I don't think he goes to t...	+44 7700 552...	HAM	2.4%	66ms	sha256:7ff0c...
15:32:23	"Your order #48217 has shipp...	ACME-SHIP	UNCERTAIN	2.0%	81ms	sha256:14980...
15:32:23	"WINNER!! As a valued custo...	PRZ-WIN	SPAM	98.4%	51ms	sha256:a3b44...
15:32:23	"Congratulations! You have b...	+1 302 555 0...	SPAM	91.8%	47ms	sha256:164ea...
15:32:23	"Reminder: Team standup mo...	+44 7911 223...	HAM	3.1%	76ms	sha256:b003f...
15:32:23	"Nah I don't think he goes to t...	+44 7700 552...	HAM	2.4%	54ms	sha256:f4615...

Showing the most recent 18 classifications · message text shown here is for analyst review only — never written to permanent storage (FR10)



Quarantine

Spam is held, never deleted — 30-day false-positive recovery window (FR3)



Release SLA: < 1 business hour

Held 128

Released 43

Auto-expired 12

Search by sender ID...

MESSAGE PREVIEW	SENDER ID	CONFIDENCE	HELD SINCE	ACTIONS
"FINAL NOTICE: Your Acme Health Plan renewal is due. Click to renew before coverage lapses: renew.acmehealth.co/r4 "	ACME-HEALTH	87.2%	2 days ago · 28d left	Release Confirm spam
"Your OTP for login is 482911. Valid for 10 minutes. Do not share this code with anyone."	OTP-SECURE	85.9%	6 hours ago · 30d left	Release Confirm spam
"URGENT: Action required — payroll direct deposit details need verification before Friday's pay run: payroll.acme-corp.io/verify "	HR-PAYROLL	91.3%	1 day ago · 29d left	Release Confirm spam
"Congratulations on completing your certification! Claim your digital badge and certificate: badges.greatlearning.io/claim/8f23 "	GL-CERTS	86.0%	3 days ago · 27d left	Release Confirm spam
"Cash-back reward of \$42.50 has been credited to your Acme Rewards account. Redeem before it expires: rewards.acme.com/redeem "	ACME-REWARDS	88.7%	5 days ago · 25d left	Release Confirm spam
"WIN tickets to the regional sales kickoff! Reply YES by Friday to enter the team prize draw."	SALES-EVENTS	85.1%	1 day ago · 29d left	Release Confirm spam



Review queue

Flow 5 · 40–60% confidence messages, routed to a human instead of auto-blocked



SLA: label within 4 business hours

Awaiting a label

14

Avg. time in queue

1h 12m

Approaching SLA (< 30m)

2

Share of total traffic

0.31%

Trigger for retraining: > 5%

Queue — oldest first

Click a message to review and label it

- ACME-SHIP

0h 19m left

"Your order #48217 has shipped — track it here: trk.acme-deliveries.co/4q1z. Reply STOP to o...

52.0% confidence
- SUB-NOTIFY

1h 55m left

"Final reminder: your subscription renews tomorrow at £9.99/mo. Manage or cancel anytime i...

47.5% confidence
- PARCEL-NOW

3h 22m left

"You've got a delivery attempt today between 1–3pm. Reschedule or confirm a safe place: [dlv.r...](#)

55.3% confidence
- OPTIFIRST

2h 48m left

"Reminder: your annual eye test is due — book online or call the practice to schedule: [optifirst....](#)

44.2% confidence
- VERIFY-CODE

3h 38m left

"Your verification code is 552190. This code expires in 5 minutes. Never share it with anyone."

58.9% confidence
- TRAINING-Q3

1h 10m left

"Limited slots remaining for the Q3 product training — reserve yours before Friday: [training.ac...](#)

49.6% confidence

🔍 UNCERTAIN · 52.0% CONFIDENCE

Time remaining in SLA

0h 19m

sha256:7c41af...b209

FROM ACME-SHIP

"Your order #48217 has shipped — track it here: trk.acme-deliveries.co/4q1z.
Reply STOP to opt out."

WHY THE MODEL IS UNSURE

Confidence sits inside the 40–60% band — the message contains terms that appear often in *both* spam and legitimate traffic, so the Decision Tree can't separate it confidently.

TRACK

LINK

REPLY STOP

✓ Label as ham — deliver normally

🛡️ Label as spam — quarantine

Your label is applied immediately and added to the next retraining run (Flow 5).

[← Live feed](#)

Message detail & explainability

Every classification is logged with the evidence behind it (§4.4)

SPAM · QUARANTINED

sha256:7c41af9e2b...b209



"WINNER!! As a valued customer you have been selected to receive a £900 prize reward! To claim call 09061701461. Code KL341."

98.4%

model confidence

From PRZ-WIN · received Jun 8, 2026 at 14:37:02 · routed to quarantine

Classified as spam because

Top contributing TF-IDF terms, ranked by weight in the Decision Tree's decision path

"WINNER"

weight 0.42

"CLAIM"

weight 0.38

"PRIZE"

weight 0.31

SENTIMENT SIGNAL (SENTIWORDNET)

Score **+0.81** — matches the bimodal urgency/reward pattern typical of spam (corpus avg. **+0.17**)

Audit trail

Logged per FR2 — message hash, label, confidence, timestamp; no raw text persisted (FR10)

**Received via SMS gateway**

14:37:02.011 · sender PRZ-WIN

**Preprocessed**

14:37:02.014 · lowercase → punctuation removed → stop words removed → tokenised (+3ms)

**TF-IDF vectorised**

14:37:02.026 · mapped to 5,572-document feature matrix (+12ms)

**Scored by Decision Tree v2.4**14:37:02.054 · label **spam**, confidence **98.4%** (+28ms)**Routed to quarantine**

14:37:02.071 · held for 30 days, recipient not notified (+17ms)

**Logged to audit trail**

Review actions

An override is logged as an analyst correction and queued for the next retraining run.

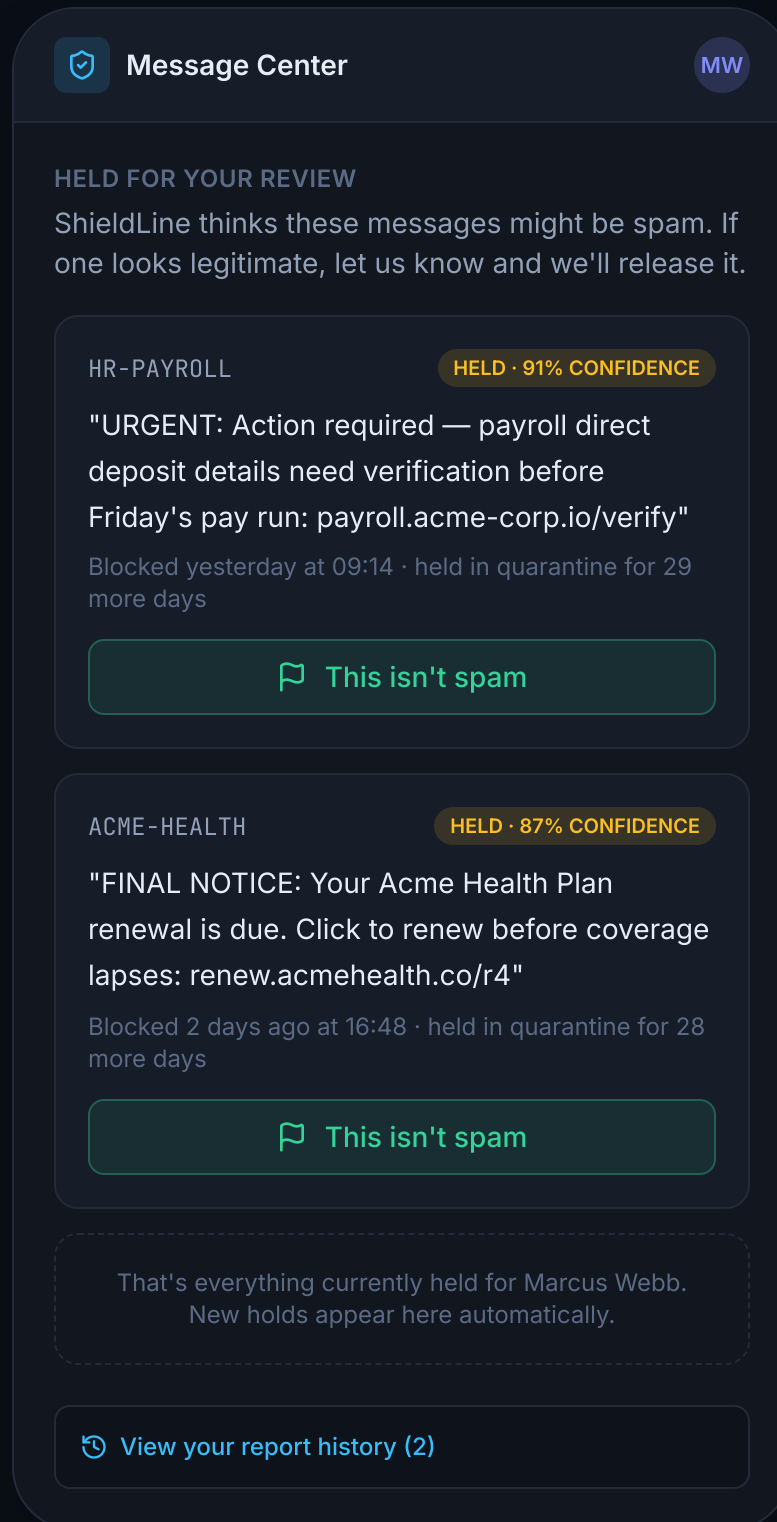
- ✓ Confirm classification
- 📁 Override — release to inbox
- 📝 Add analyst note

Message metadata

Sender ID	PRZ-WIN
Channel	SMS · enterprise gateway
Model version	decision-tree-v2.4
Total latency	68ms (P99 target <100ms)
Stored content	Hash only — GDPR/CCPA
Retention	29 days remaining

[View in quarantine queue](#) →

Flow 2 — when ShieldLine blocks a legitimate message, the recipient can flag it from this lightweight portal in under 3 taps (Story 2 acceptance criteria).



Persona: Marcus Webb, Business Operations Manager — Story 2: "I want to report the misclassification and have the message released within 1 business hour."



Alerts & campaigns

Flow 4 · spike detection within 15 minutes, weekly reporting (FR7)

Email weekly report

Active alerts 2 open



Spam rate spike — +24% above 7-day rolling average

CRITICAL

INVESTIGATING



Triggered Jun 8, 2026 at 14:32 · assigned to Priya Nandakumar · fired 6 minutes after threshold breach (target < 15 min)

SPAM RATE VS. 7-DAY AVERAGE

08:00

10:00

12:00

13:00

14:00

15:00

Spam rate climbed from a baseline of 13.1% to **16.2%** within two hours — consistent with a coordinated campaign rather than normal variance.

SAMPLE MESSAGES DRIVING THE SPIKE

"CASH prize of £5000 is waiting! Text WIN to 80082 now to claim before midnight..." · CASH-LINE

"Congratulations! You've been selected for a FREE flagship phone. Click to claim..." · +1 302 555 0142

"URGENT! Your Mobile No. has won £2000 Bonus Caller Prize. Call 09066362231..." · +44 7700 900112

RECOMMENDED ACTIONS

- ☒ Confirm campaign pattern against top spam phrases (CASH, FREE, WIN/WINNER)
- ☐ Review related senders for allowlist/blocklist action in Settings
- ☐ Flag the affected phrases for priority inclusion in next month's retraining corpus

Acknowledge

Mark resolved



New spam pattern detected — "claim your reward" phrase surge

MEDIUM



INVESTIGATING

Triggered Jun 7, 2026 at 09:14 · assigned to Devon Reyes · 41 occurrences in 24 hours (vs. 6 the week before)

Recently resolved



False positive rate approached threshold — peaked at 6.3% (target $\leq 6.5\%$)

Resolved Jun 6, 2026 · root cause: a legitimate appointment-reminder campaign sharing vocabulary with spam · added 40 examples to allowlist review

Low

Weekly security report — Jun 1–7, 2026

Auto-generated every Monday and emailed to the security distribution list (Story 4)



Export CSV



View full report

Messages classified

1.21M

Spam rate

13.8%

Precision / recall

93.7% /
91.2%

New spam phrases

"claim your
reward",
"verify
account
now"



Model & retraining

Flow 3 · monthly retraining with an accuracy gate before any promotion (FR5)

Production: decision-tree-v2.4



Decision Tree (before pruning) — v2.4 IN PRODUCTION

Recommended model per §5.2 — best balance of recall (catching spam) and precision (not over-blocking) for a security use case. Promoted Jun 1, 2026.

Test accuracy

96.59%

Test precision (spam)

93.50%

Test recall (spam)

91.50%

Trained on

5,572 msgs

Full model comparison (§5.1)

Four candidates evaluated on a 70/30 train/test split of the 5,572-message corpus (4,825 ham · 747 spam)

MODEL	TRAIN ACC.	TEST ACC.	TEST RECALL	TEST PRECISION	VERDICT
Decision Tree (before pruning)	97.22%	96.59%	91.50%	93.50%	Selected — best balance
Decision Tree — pruned	96.75%	94.79%	89.90%	88.18%	Lower precision
Random Forest (before pruning)	95.83%	95.42%	82.89%	97.49%	Misses 17% of spam
Random Forest — pruned	97.49%	94.70%	85.31%	90.78%	Lower recall

Why this trade-off is acceptable: a 93.5% precision means roughly 6.5% of legitimate messages may be quarantined — but because spam is held (not deleted) and false positives are released within 1 business hour, the security upside of catching 91.5% of spam outweighs that inconvenience cost (§5.2).

Last retraining run — Jun 1, 2026

✓ Passed accuracy gate



Pulled training data

1,842 confirmed feedback corrections + last 30 days of classified messages → corpus grew to 7,414 labelled examples



Retrained Decision Tree on expanded corpus

Completed in 2h 41m · within the 4-hour monthly window (§4.4 Performance)



Evaluated against accuracy gate

Precision (need ≥ 93%)

93.8% ✓

Recall (need ≥ 90%)

91.2% ✓



Promoted to production as v2.4

Approved by Devon Reyes (ML Engineer) · v2.3 archived with rollback capability — human approval required per §4.3

Pipeline schedule

Next scheduled run

Jul 1, 2026 · 04:00 UTC

Promotion gate

precision ≥ 93% & recall ≥ 90%

If gate fails

Hold for manual review, alert ML team

▶ [Trigger retraining run now](#)

Version history

Every promoted (and rejected) version is retained with rollback capability

v2.4

ACTIVE

Promoted Jun 1, 2026 · precision 93.8% · recall 91.2%

Current production model

v2.3

ARCHIVED

Promoted May 1, 2026 · precision 93.2% · recall 90.6%

[Request rollback](#)

v2.2

ARCHIVED

Promoted Apr 1, 2026 · precision 92.9% · recall 90.1%

[Request rollback](#)

v2.1-rc

REJECTED

Evaluated Mar 1, 2026 · recall 89.3% — below the 90% gate; v2.0 retained, ML team alerted



Settings & governance

§4.3 · changes that affect classification behavior require a second approver

General

Classification thresholds

Sender allowlist

API keys

Team & roles

Organization

Organization name

Acme Mobile

Time zone

UTC (Coordinated Universal Time)

Deployment mode

Per the go-to-market plan in §6 — controls whether spam is blocked or only measured

☐ Shadow mode

Classify and log everything; deliver all messages — used to tune thresholds before blocking

☒ Live — quarantine enabled

High-confidence spam ($\geq 85\%$) is held; feedback loop and human review are active

Data & retention

Quarantine retention window

30 days

Message content storage

 In-memory only — SHA-256 hash retained (FR10)